

Nakka Chakradhar

Work Experience

2026 Jan -
Current

Senior ML Engineer; The Real Brokerage, USA (India - Remote)

- Built **leo-vectors**, a Ray Serve GPU embedding service (SigLIP v2, Gemma, DINOv2 dedupe) with realtime / batch scheduling
- Shipped enhanced image search on 100+M images - Gemini VLM captioning, OpenSearch property embeddings, vector reranking, buyer ranking
- Built agent workflows in **Ari** / **Hawkeye** - LLM-judged re-engagement outreach, agents-as-code publishing, agent-run verification loop, Kafka-triggered tasks

2023 Oct -
2025 Dec

AI Research Engineer; DeepReel, London (India - Remote)

- Developed and fine-tuned FLAME-based 3D facial tracking, lipsync models, neural textures and neural rendering pipelines for avatar training and inference
- Built video processing workflows including frame extraction, facial detection (MediaPipe, FAN-68), person centering, and codec optimization for inference
- Implemented perceptual losses for improved facial texture in neural rendering; researched Gaussian Splatting for enhanced facial expressions
- Architected **Genie**, an agentic video-generation platform orchestrating LLMs (LangFuse, BAML) for end-to-end video production - script generation, scene planning, B-roll search / generation
- Integrated stock media search, Google search APIs, and FAL-based generative models to create explainer videos at scale

2022 Dec -
2023 Sep

AI R&D Lead Engineer; Frinks AI, Bangalore

- Led R&D for platform components enabling high-throughput visual inspection
- Developed object detection, dimensioning, segmentation and anomaly detection models to identify product defects at scale, processing 120 products per minute

2020 Sep -
2022 Dec

Graduate Software Engineer; Jaguar Land Rover, Bangalore

- Worked as ADAS Engineer on 3D object detection in Lidar domain and path planning using PyTorch

Education

2016-2020

BTech, Electrical Engineering; Indian Institute of Technology, Hyderabad

CGPA : 9.00

Projects

Flux LoRA Fine-tuning

Experimented with fine-tuning Flux, Qwen-Image LoRAs for character consistency, virtual try-on and style transfer applications.

Supervised Deep Learning for MIMO Precoding : [Link](#)

Demonstrated that supervised learning with noisy back-propagation could achieve Shannon capacity for channel coding without reinforcement learning. Published in IEEE.

Lung Tumor Segmentation

Segmentation of lung tumors on DICOM images for IEEE VIP-CUP 2018 competition.

Gait Recognition with Keras

Cascaded HumanPoseNN and GaitNN for deep learning-based gait recognition.

Gesture Recognition (Inter IIT Tech Meet 2017)

Developed glove-based gesture recognition module capturing hand movements in 3D space for DRDO's 'Soldier Support' problem.

Technical Skills

ML Frameworks: PyTorch, TensorFlow, Keras

Generative AI & Video Tools: Stable Diffusion, Flux (fine-tuned LoRAs), FAL, ComfyUI, Swarm, HuggingFace

Video & Image Processing: FFmpeg, OpenCV, MediaPipe

LLM & Agentic Systems: LangChain, LangFuse, BAML (structured outputs)

Computer Vision: Object Detection, Segmentation, Tracking, Facial Recognition (MediaPipe, FAN-68)

Cloud Platforms: GCP, Azure

Focus Areas: Generative Models, Image and Video Processing & Generation, Computer Vision, VLMs/LLMs, Diffusion Transformers

Achievements and Leadership

- First prize in GDD Hackathon at JLR for 360-degree vision system using computer vision
- Teaching Assistant under Dr. Sushmee Badhulika and Dr. Sumohana Chennappayya (Professors, Department of Electrical Engineering, IIT Hyderabad)

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